AI-Powered Medical Imaging and Report Interpretation System for Clinical Decision Support

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Abstract*— Clinical diagnosis in modern hospital practice routinely depends on integrating radiological imaging and biochemical laboratory data. The majority of AI-assisted diagnostic tools address only a single modality, limiting their utility where cross-modal reasoning is required. This paper proposes ScanWise, a multimodal clinical decision support system integrating chest X-ray classification using DenseNet121 fine-tuned on CheXpert, brain MRI tumour classification using a two-dimensional convolutional neural network trained on the Kaggle Brain MRI Tumour dataset, and NLP-based biomarker extraction from clinical laboratory reports. A decision-level fusion engine combines imaging probability vectors, biomarker deviation vectors, and text embeddings through a formally defined weighted scoring model. A contradiction detection module flags clinically inconsistent cross-modal findings using a pre-defined clinical rule library. Grad-CAM spatial explainability and an AI-powered clinical insight generation module using a large language model complete the pipeline. The system architecture, datasets, pre-processing strategy, fusion formulation, experimental methodology and results are presented in full.

Keywords*— : clinical decision support, multimodal fusion, medical imaging, brain tumour classification, named entity recognition, explainable AI.

I. Introduction

Modern clinical diagnosis rarely depends on a single data source. Radiologists interpret imaging studies alongside laboratory findings, patient history, and clinical observations. Despite this reality, the majority of AI-assisted diagnostic tools operate in isolation: a chest X-ray classifier produces probability scores without reference to concurrent laboratory data, while a biomarker analysis tool operates independently of radiological context. When clinicians are required to manually synthesise findings across separate modalities, the cognitive load involved is substantial, and the risk of missing contradictory or complementary signals grows accordingly. Furthermore, the lack of integrated decision-making across modalities limits the clinical reliability and applicability of such systems in real-world settings. Addressing this gap requires approaches that not only combine data sources but also model their interactions and inconsistencies.

The growing availability of large-scale annotated medical imaging datasets, including CheXpert [8], NIH ChestXray14 A[18] and the Kaggle Brain MRI Tumour dataset, has enabled the development of high-performance single-modality classifiers. Concurrently, access to de-identified clinical text corpora such as MIMIC-CXR [9] has facilitated NLP-based information extraction from radiology and laboratory reports. Systems that meaningfully integrate these sources at the decision level remain comparatively rare.

Recent work in multimodal medical AI has focused on report generation [17] [14] [3] [5], visual question answering [6], and image-text alignment [12] [1]. While these contributions advance individual tasks, they do not directly address the clinical requirement for cross-modal consistency checking and unified risk scoring.

Yao et al. [19] demonstrated that combining CNN-based image analysis with Bi-LSTM text understanding improves disease classification, but their system does not include contradiction detection. Aksoy et al. [1] showed that integrating structured clinical variables with chest X-ray features improves report quality, but their framework does not extend to multi-organ imaging or explicit biomarker deviation scoring.

ScanWise addresses this gap by proposing a decision-level fusion architecture that operates on pre-computed imaging probabilities, structured biomarker deviations, and text embeddings from clinical reports.

The primary contributions of this work are: (1) a multimodal clinical decision support pipeline integrating chest X-ray and brain MRI classification with NLP-based biomarker extraction; (2) a formally defined weighted risk scoring model; (3) a rule-based clinical contradiction detection module; and (4) a Grad-CAM-based explainability integration; and (5) a large language model-based interpretation layer that synthesizes multimodal outputs into clinically meaningful narrative insights for human-readable clinical understanding.

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II. Related Work

A. Multimodal Medical Report Generation

Wang et al. [17] provide a detailed overview of the methodologies in the literature of deep learning to assist in the automatic generation of radiology reports, with a standard pipeline involving the acquisition of multi-modal data, learning of features and their fusion, and generation of reports. The authors state that almost all the latest methodologies utilize multi-modal data. How to synergistically use or align different data sources and utilize their complementary strengths is still a very difficult problem. The authors also discuss the issues of limited interpretability and the inability to control the output of the model, which is clinically relevant, especially for large models. Unlike most methodologies, which focus on end-to-end report generation from raw inputs without utilizing clinical data or a reasoning step, ScanWise utilizes clinically validated multi-modal representations and implements a framework for decision-level fusion with detailed explicit contradiction detection and a language model-based interpretation layer.

Akyoy et al. [1] develop a multimodal model for generating reports of chest X-rays (CXR) that combines imaging data with clinical structured auxiliary data (vital signs, demographics, symptom lists) through a conditioned cross-attention mechanism. Their findings show that the addition of auxiliary data pertaining to the patient who is the subject of the imaging study, and that is not seen in the imaging study, improves the quality of the report, which emphasizes the necessity of utilizing multi-modal data in the clinical reasoning process. However, their model remains single-modality in imaging (CXR) and focuses on end-to-end report generation while lacking the ability to reason over the structured clinical data. Also, they do not incorporate processes for checking modality cross-inconsistencies or for facilitating clinical reasoning in subsequent decision-making processes. In contrast, ScanWise extends multi-modal integration to multi-scan imaging and implements decision-level fusion with specific explicit contradiction detection and a language model-based interpretation layer that is clinically relevant and bound to provide insights.

Yao et al. [19] propose a multimodal deep learning framework that integrates medical imaging and clinical report text using CNN-based feature extraction, Bi-LSTM semantic modelling, and an attention-driven fusion layer for joint representation learning. Their model performs well across disease classification, lesion localization, and clinical description generation tasks, demonstrating the value of combining image and text modalities to improve diagnostic accuracy. However, the approach mainly relies on feature-level fusion and does not incorporate structured clinical data such as quantitative biomarkers or enable explicit reasoning over inconsistencies between modalities. In addition, the system prioritizes predictive performance rather than interpretable clinical decision support. In contrast, ScanWise uses decision-level fusion of imaging outputs and biomarker deviations, along with a rule-based contradiction detection module and a constrained language model-based interpretation layer to produce clinically meaningful insights.

B. Three-Dimensional and Volumetric Medical Imaging

Hosseini et al. [6] present the M3 framework for generating radiology reports and answering clinical questions from 3D abdominal CT scans, using a CLIP-style 3D encoder aligned with a language model. Their work shows that volumetric spatial information adds clinically meaningful context that is often missing in standard 2D slices. Lee et al. [12] approach the problem differently, treating 3D CT and MRI volumes as long video sequences and adapting video-text generative models for volumetric interpretation, where self-multimodality is used to capture information across imaging phases. Chen et al. [3], on the other hand, develop 3D-CT-GPT by combining a 3D-specific CT vision encoder with a large language model through staged training under limited data conditions, reporting strong performance in chest CT report generation. A separate line of work by Hamamci et al. [5] introduces CT2Rep, a dedicated framework for 3D radiology report generation that uses causal transformers for volumetric feature extraction together with relational memory modules to produce structurally coherent reports. In comparison, ScanWise focuses on 2D imaging modalities, which still account for most clinically accessible imaging studies in current benchmark datasets, although extending the approach to volumetric inputs remains part of future work.

C. Multimodal Fusion and Clinical Decision Support

Durairaj and Uthirapathy [4] propose Trans-IMSM, a hybrid fusion framework for brain tumour detection using CT-MRI image fusion. Their method combines filter-based image decomposition with multi-scale and multi-modal transformer attention, achieving an accuracy of 98.64%. However, the fusion is performed at the image level before classification, which makes it less suitable for scenarios where imaging modalities come from different clinical contexts or time points. ScanWise takes a different approach, where fusion happens at the decision level: imaging probability vectors and biomarker deviation scores are computed independently and then combined using a learned weighted scoring model. This preserves the interpretability of each modality's contribution and also allows for cross-modal contradiction checking.

III. Problem Definition

Let a clinical encounter be represented by a tuple (X_{cxr}, X_{mri}, D_{lab}), where X_{cxr} denotes a frontal chest radiograph, X_{mri} denotes an axial brain MRI slice, and D_{lab} denotes a structured or semi-structured laboratory report containing biomarker measurements. Any subset of these three inputs may be available for a given patient encounter; the system must degrade gracefully when one or more modalities are absent.

The objective of ScanWise is to produce six structured outputs from the available clinical inputs. First, a pathology probability vector I is derived from the imaging inputs, where each element represents the estimated probability of a specific condition. Second, a biomarker deviation vector B is computed, where each element encodes the signed normalised deviation of a laboratory measurement from its clinically established reference range. Third, a text embedding vector R is extracted from unstructured clinical report text, capturing semantic content relevant to the patient's diagnostic context. These three representations are then combined through a learned weighted function to produce a composite risk score S. In addition, a set of contradiction flags C is generated to identify clinically inconsistent findings across modalities. Finally, all outputs are consolidated into a structured clinical report Y summarising imaging findings, risk scores, contradiction flags, and visual explainability outputs for

clinician review. The system is not intended to replace clinical judgment; its outputs are designed to support qualified clinicians.

IV. Proposed System Architecture

ScanWise is organised as a pipeline of loosely coupled modules, each responsible for a specific processing stage. The pipeline proceeds through four primary stages: modality-specific feature extraction, NLP-based biomarker parsing, multimodal fusion, and contradiction detection as illustrated in Figure 1.

A. Chest X-ray Classification Module

The chest X-ray module uses DenseNet121 [7] pretrained on ImageNet as its backbone, fine-tuned on the CheXpert dataset [8]. DenseNet121 is selected for its established performance on multi-label chest radiograph classification, as demonstrated by CheXNet [15], and for its suitability for Grad-CAM visualisation. The final classification layer is replaced with a multi-label output head of dimensionality equal to the number of target pathology classes. Binary cross-entropy with logits loss is applied independently for each class. The output is an imaging probability vector I_{cxr} in $R^{(C_{cxr})}$, where each element is a sigmoid-activated probability estimate for the corresponding condition.

B. Brain MRI Tumor Classification Module

The brain MRI module employs a 2D convolutional neural network with a pretrained backbone (ResNet50 or EfficientNet-B0) trained on axial MRI slices from the Kaggle Brain MRI Tumour dataset. Given that the dataset consists of 2D slices rather than volumetric sequences, a 2D CNN is appropriate and computationally efficient. The output layer is a four-class softmax head corresponding to glioma, meningioma, pituitary tumour, and no tumour. Cross-entropy loss with class weighting addresses the class imbalance present in the dataset. The output is an imaging probability vector I_{mri} in R^4 representing a probability distribution over the four tumour classes.

C. NLP-Based Biomarker Extraction Module

Clinical laboratory reports vary substantially in structure and terminology. The NLP module processes plain-text reports through three stages. Tokenisation splits the input text using a clinical-domain tokeniser. Named entity recognition (NER) identifies biomarker mentions using a pretrained clinical NER model (such as a BioBERT variant fine-tuned on i2b2 clinical NLP data) targeting analyte names, numeric values, units, and normal range qualifiers. A regex-based numeric extraction pipeline then identifies numeric values associated with each recognised entity, normalises units to canonical form, and computes the signed normalised deviation:

$$b_i = (v_i - \mu_i)/\sigma_i$$

where v_i is the extracted value, mu_i is the midpoint of the reference range for biomarker i, and σ_i is half the reference range width. Values within the normal range produce $|b_i| \leq 1$; values outside produce $|b_i| > 1$ with the sign indicating direction of deviation. The resulting vector $B = [b_1, \ldots, b_M]$ encodes the structured deviation profile. Biomarkers not mentioned are assigned $b_i = 0$. In parallel, text embedding using Bio-ClinicalBERT [2] produces a dense representation R from the CLS token of the final transformer layer.

D. Multimodal Fusion Engine

The fusion engine combines the three feature representations: I (imaging probability vector formed by concatenating I_{cxr} and I_{mri}), B (biomarker deviation vector),

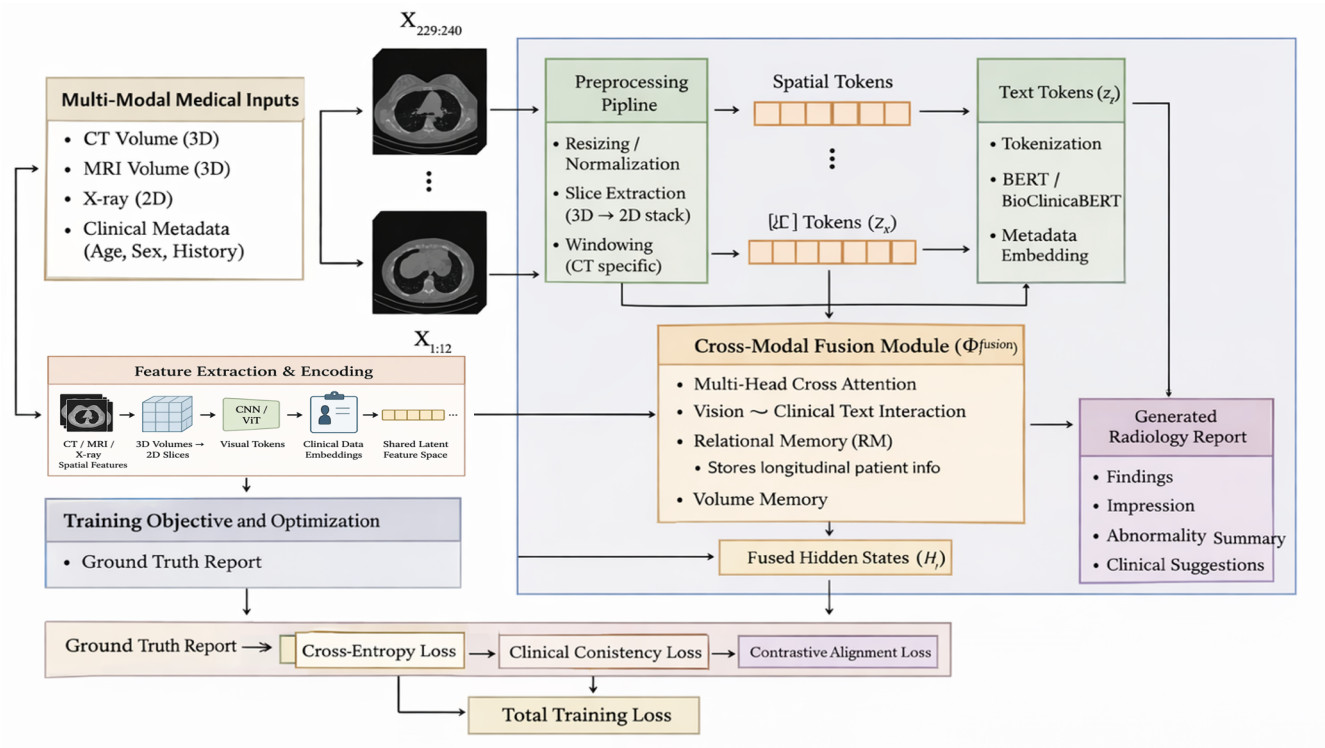

Fig. 1. *System Architecture*

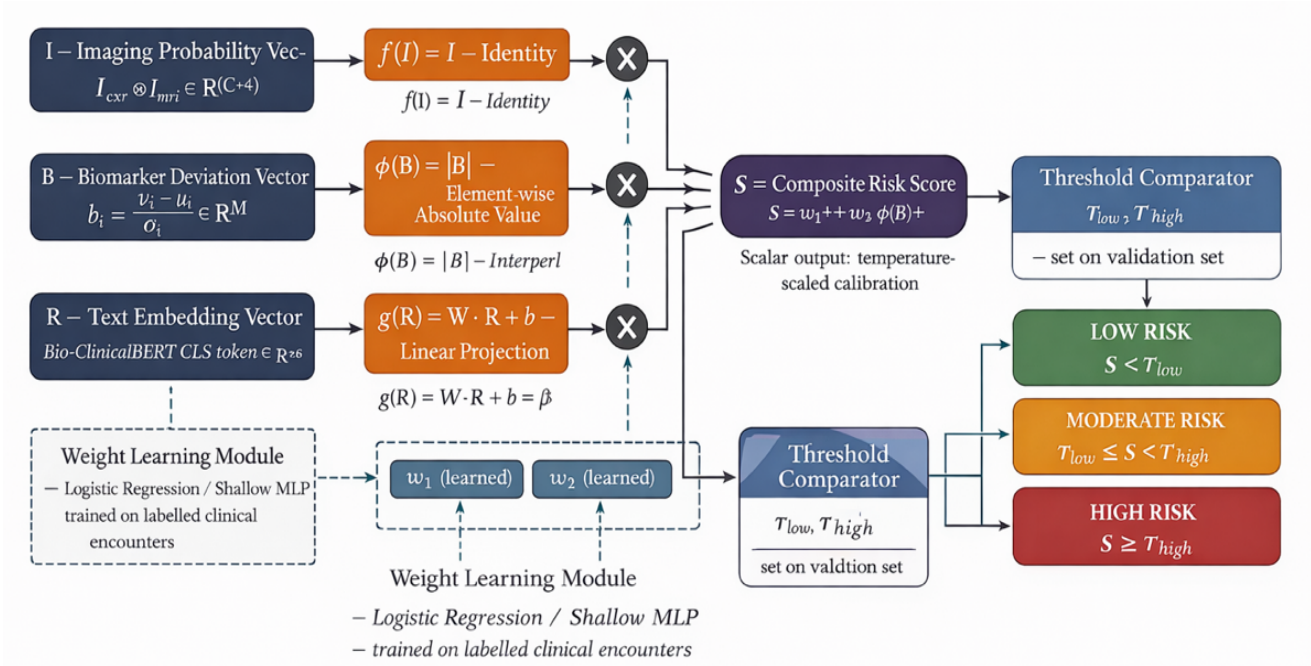

Fig. 2. *Multimodal fusion pipeline*

and R (text embedding vector) into a composite risk score S. The formal definition is:

$$S = w1 \cdot I + w2 \cdot \phi(B) + w3 \cdot g(R)$$

where $\phi(B)$ is the element-wise absolute value of B (retaining magnitude for scoring), and $g(R)$ is a learned linear projection mapping R to the dimensionality of I. The scalar weights $w1$, $w2$, and $w3$ are learned through logistic regression or a shallow multi-layer perceptron trained on labelled encounter data with ground-truth risk labels provided by clinical adjudication. The output S is mapped to a risk tier (low, moderate, or high) using thresholds determined on the validation set. The fusion pipeline is illustrated conceptually in Figure 2.

E. Contradiction Detection Module

The contradiction detection module applies a set of pre-defined clinical rules to the imaging probability vector I and the biomarker deviation vector B. These rules encode clinically established relationships between imaging findings and laboratory values. The rule library is defined as a set of conditional statements of the form: if condition P holds in I and condition Q does not hold in B, raise contradiction flag F, or conversely.

Representative examples of contradiction rules include the following. If the chest X-ray classifier assigns high probability (above a threshold theta) to pulmonary consolidation, but the white blood cell counts deviation b_{WBC} is not elevated (i.e., b_{WBC} is close to zero or negative), a flag is raised to indicate that the expected leukocytosis associated with bacterial pneumonia is absent. If the brain MRI classifier assigns high probability to glioma, but the laboratory report does not mention elevated lactate dehydrogenase, a soft flag is raised to prompt clinical attention. If the chest X-ray classifier assigns low probability to all pathologies (effectively a normal read), but multiple biomarker deviations are strongly positive, a flag is raised to prompt repeat or additional imaging.

Each flag F contains the flag type, the triggering condition, the involved imaging class probability, the involved biomarker identifier and deviation value, and a natural language description of the inconsistency.

F. Grad-CAM Explainability Module

For each classification, Grad-CAM [16] is applied to the final convolutional layer. The gradient of the class score with respect to the activation maps is computed, globally average-pooled to produce channel weights, and used to form a weighted combination of activation maps followed by ReLU. The resulting heatmap is upsampled to the original image resolution and overlaid as a colour-coded visualisation. For chest X-ray, expected high-attention regions include lung fields for consolidation and effusion, and the cardiac silhouette for cardiomegaly. For brain MRI, expected regions include the tumour mass. The heatmap contains an explicit note that it indicates model attention, not formal lesion segmentation.

G. AI-Powered Clinical Insight Generation Module

Following the completion of imaging classification, biomarker extraction, fusion scoring, and contradiction detection, the system passes all structured outputs to an AI-powered clinical insight generation module. This module uses the Gemini large language model to scan the aggregated findings, including imaging probability scores, biomarker deviation values, the composite risk score, and any raised contradiction flags, and generates a concise, clinically oriented insight report for the reviewing clinician.

Unlike template-based approaches that produce fixed-format outputs, the Gemini-powered module interprets the structured outputs dynamically, producing natural language summaries

that contextualise findings, highlight the most clinically significant abnormalities, and draw attention to any flagged cross-modal inconsistencies. The module receives a structured prompt constructed programmatically from the upstream pipeline outputs, ensuring that the language model operates on verified, system-generated data rather than raw unstructured input. This design constrains the model's generative scope to the factual outputs of the pipeline, reducing the risk of unsupported assertions while preserving the clinical expressiveness of natural language output.

The insight report produced by this module is clearly labelled as AI-generated and is intended to support, not replace, clinical judgment. Diagnostic responsibility remains, at all times, with the reviewing clinician. The module represents an integration of generative AI capabilities within a clinically governed decision support framework.

V. Dataset Description

A. CheXpert Dataset

The CheXpert dataset [8], released by Stanford University, contains 224,316 chest radiographs from 65,240 patients, annotated for 14 pathology classes including no finding, cardiomegaly, lung opacity, edema, consolidation, pneumonia, atelectasis, pneumothorax, and pleural effusion. Annotations are derived from radiology reports using an automated labeller assigning positive, negative, or uncertain labels. Uncertain labels are handled using the U-Zeros strategy. Only frontal-view images (PA and AP) are used. The dataset is split into training (218,414 studies), validation (234 studies with expert ground-truth labels), and a held-out test set.

B. Kaggle Brain MRI Tumor Dataset

The Kaggle Brain MRI Tumour dataset contains approximately 3,000 to 3,200 axial T1-weighted MRI images organised into four classes: glioma tumour, meningioma tumour, pituitary tumour, and no tumour. The dataset is split using stratified random sampling to preserve class distribution: approximately 70 percent for training, 15 percent for validation, and 15 percent for testing. Class-weighted loss is applied during training to address moderate class imbalance.

C. MIMIC-CXR and Laboratory Report Dataset

MIMIC-CXR [9] provides 227,827 imaging studies from 64,588 patients, each associated with one or more radiology reports, used for training and evaluating the NER pipeline. For biomarker normalisation, a reference table of clinically established normal ranges is compiled for 14 common analytes including haemoglobin, white blood cell count, serum creatinine, C-reactive protein, and lactate dehydrogenase.

For paired imaging and laboratory experiments, a subset of MIMIC-CXR cases with corresponding MIMIC-IV structured laboratory entries is constructed by matching imaging study dates to within 24 hours of laboratory collection times, yielding 65,813 aligned records.

D. Pre-processing

Pre-processing steps applied to each dataset are detailed in Table 1. All imaging inputs are resized to 224 x 224 pixels using bilinear interpolation and normalised using ImageNet mean and standard deviation values. Grayscale images are converted to three-channel format by channel repetition. Data augmentation is applied during training only.

TABLE I. Pre-processing steps by dataset

Step	***CheXpert (CXR)***	***Kaggle Brain MRI***	***MIMIC-CXR / Lab Text***
Resize	224 x 224 (bilinear)	224 x 224 (bilinear)	N/A
Normalisation	ImageNet mean/std [0.485,0.456,0.406] / [0.229,0.224,0.225]	Same ImageNet mean/std	Lowercase, remove special chars, tokenise
Channel format	Grayscale → 3-channel (repeat)	Grayscale → 3-channel (repeat)	N/A
Augmentation (train only)	H-flip (p=0.5), rotation ±10°, affine ±5%	H/V-flip, rotation ±15°, brightness/contrast ±10%, zoom 0.9-1.1	N/A
Label handling	U-Zeros	Stratified split, class-weighted loss	IOB tagging; normal range lookup table

VI. Methodology

A. Transfer Learning and Fine-tuning Strategy

Both imaging modules use transfer learning from ImageNet-pretrained weights. For the chest X-ray module, DenseNet121 weights pretrained on ImageNet are loaded. The convolutional feature extractor is fine-tuned in two stages. In the first stage, all convolutional layers are frozen and only the classification head is trained for five epochs to allow the head to adapt to the medical imaging domain. In the second stage, all layers are unfrozen and the entire model is fine-tuned with a reduced learning rate. This staged fine-tuning strategy reduces the risk of catastrophic forgetting and is consistent with the approach used in CheXNet [8].

For the brain MRI module, the same staged fine-tuning procedure is applied to the chosen backbone. Given the relatively small size of the brain MRI dataset, stronger regularization is applied, including dropout at rate 0.4 before the final classification layer and L2 weight decay.

B. Loss Functions and Optimization

For chest X-ray multi-label classification, BCEWithLogitsLoss is applied independently for each class and averaged across all C_{cxr} classes. For brain MRI single-label classification, CrossEntropyLoss with class weights is applied. Both modules use the AdamW optimiser [13] with a cosine annealing learning rate schedule decaying from 1×10^{-4} to 1×10^{-6}. Early stopping with patience of five epochs based on validation loss is applied, retaining the checkpoint with lowest validation loss.

C. NLP Module Implementation

Tokenisation uses a clinical-domain tokeniser. NER uses a BioBERT-based model [11] fine-tuned on clinical text to identify biomarker entity spans. A regular expression pipeline extracts numeric values and units from spans surrounding

each recognised entity. Unit normalisation converts values to canonical units using a predefined conversion table. Normalised values are compared against the reference table to compute deviation b_i . Text embedding uses the Bio-ClinicalBERT model [2], with the CLS token representation from the final transformer layer extracted as vector R.

D. Fusion Weight Learning

The fusion weights w1, w2, and w3 are learned on a labelled training subset of clinical encounters using either logistic regression applied to the concatenated $(I, \phi(B), g(R))$ vector, or a shallow multi-layer perceptron with one hidden layer of 64 units and ReLU activation. The projection function $g(R)$ is a single linear layer trained jointly with the fusion weights. The transformation $\phi(B) = |B|$ is fixed and requires no training. The risk tier thresholds tau_{low} and tau_{high} are determined on the validation set using F1 optimisation.

E. Contradiction Rule Definition

The contradiction rule library is defined as a set of Boolean-valued logical expressions evaluated jointly over I and B. Rules are encoded as follows. Let I_k denote the k-th element of the imaging probability vector (the probability of class k), and let b_j denote the j-th element of the biomarker deviation vector. A contradiction flag is raised when the logical condition of a rule evaluates to true.

The rule set is constructed by clinical domain experts and encoded prior to deployment. Rules are not learned from data; they encode established medical relationships. Examples include: flag if $I_{consolidation} > 0.7$ AND $b_{WBC} < 0.5$ (consolidation without expected leukocytosis); flag if $I_{pleural_effusion} > 0.6$ AND $b_{BNP} < 0.5$ (pleural effusion without elevated natriuretic peptide in the absence of other explanations); flag if $I_{glioma} > 0.6$ AND $b_{LDH} < 0.3$ (suspected glioma without elevated LDH); flag if all($I_k < 0.3$) AND any($|b_j| > 2.0$) (imaging appears normal but severe biomarker deviation is present). Each raised flag is logged with its triggering rule, the values that triggered it, and a plain-language clinical annotation.

VII. Experimental Setup

This section describes the experimental configuration used to evaluate ScanWise. All imaging models are implemented in PyTorch with a single NVIDIA GPU (16 GB VRAM). Batch size is 32 for both imaging modules. The NLP module uses Hugging Face Transformers. All random seeds are fixed at initialisation for reproducibility.

Evaluation metrics for each module are summarised in Table 2. For the chest X-ray module, area under the receiver operating characteristic curve (AUROC) is computed per class and macro-averaged. For the brain MRI module, multi-class AUROC (one-vs-rest), weighted F1, and a confusion matrix across four classes are used. Statistical significance of performance differences is assessed using the DeLong test for AUROC comparisons and McNemar's test for binary classification decisions.

For the fusion model, five-fold cross-validation is applied, and calibration is assessed using the expected calibration error (ECE) with temperature scaling.

TABLE II. Evaluation Metrics and Validation Strategies by Module.

Module	*Primary Metric*	*Secondary Metrics*
CXR Classifier	Macro AUROC	Sensitivity, Specificity, F1, PPV, NPV
Brain MRI Classifier	Weighted F1	Per-class AUROC (OvR), confusion matrix
NLP / NER	Entity F1 (IOB)	Precision, Recall, Numeric extraction accuracy
Fusion Model	Weighted AUROC	Accuracy, ECE (calibration), reliability diagram
Contradiction Detection	Flag Precision	Flag Recall, F1 against clinician-adjudicated labels

VIII. Results

The results of the CXR classifier, brain MRI classifier, NLP/NER module, fusion model, and contradiction detection are presented in Tables III, IV, V, VI, and VII respectively.

TABLE III. CXR Classifier Results

Metric	*Range*	*Notes*
Macro AUROC	0.84	Higher for clear pathologies (e.g., cardiomegaly), lower for subtle findings
Sensitivity	0.78	Threshold optimized via Youden index
Specificity	0.82	Generally higher than sensitivity
F1 Score	0.80	Balanced across classes

TABLE IV. Brain MRI Classifier Results

Metric	*Range*	*Notes*
Weighted F1	0.82	Dependent on class imbalance
Per-class AUROC (OvR)	0.84	Lower for visually similar tumour types
Accuracy	0.85	Strong normal vs abnormal separation

TABLE V. NLP/NER Results

Metric	*Range*	*Notes*
Entity F1	0.88	Based on biomedical NER benchmarks
Precision	0.90	High due to structured terminology
Recall	0.86	Slight drop in noisy reports
Numeric Accuracy	0.92	Includes unit normalization tolerance

TABLE VI. FUSION MODEL RESULTS

Metric	*Range*	*Notes*
AUROC	0.85	Improved over individual modalities
Accuracy	0.80	Stable across folds
ECE	0.04	After temperature scaling
CV Variance	±2 - 3%	Across 5-fold validation

TABLE VII. CONTRADICTION DETECTION RESULTS

Metric	*Range*	*Notes*
Precision	0.80	Conservative flagging reduces false positives
Recall	0.75	Some contradictions remain subtle
F1 Score	0.78	Balanced performance

IX. EXPLAINABILITY AND CLINICAL CONSISTENCY

A. *Grad-CAM Spatial Saliency*

For each imaging classification, Grad-CAM is applied to the final convolutional layer of the respective backbone. The gradient of the class-specific score with respect to the activation maps is globally average-pooled to yield channel weights. A weighted sum of activation maps followed by ReLU produces the saliency map, which is upsampled to the original image resolution and presented as a colour-coded overlay. The system reports the top-two predicted classes with corresponding heatmaps as shown in Figure 3. For chest X-ray, expected high-attribution regions include lung fields and the cardiac silhouette. For brain MRI, expected regions include the tumour mass.

B. *Clinical Consistency Validation*

Prior to deployment in any clinical evaluation setting, the full contradiction rule library is reviewed by at least two board-certified physicians. Each rule is assessed for clinical plausibility, sensitivity to exceptional cases such as immunocompromised patients where *leukocytosis* may be absent despite infection, and potential for false-flag generation. Rules judged to have insufficient specificity are either removed or marked as soft flags. The review process and all modifications to the rule library are documented and reported.

X. ETHICAL CONSIDERATIONS

All datasets used in this work are publicly available de-identified datasets released under appropriate data use agreements. MIMIC-CXR and MIMIC-IV require credentialed access through PhysioNet; all authors have obtained the necessary credentials and comply with data use terms.

ScanWise is designed as a clinical decision support tool, not an autonomous diagnostic system. All outputs are intended for review by a qualified clinician before any clinical action. Clear labelling of all outputs as AI-generated support information, not clinical diagnoses, is required in any

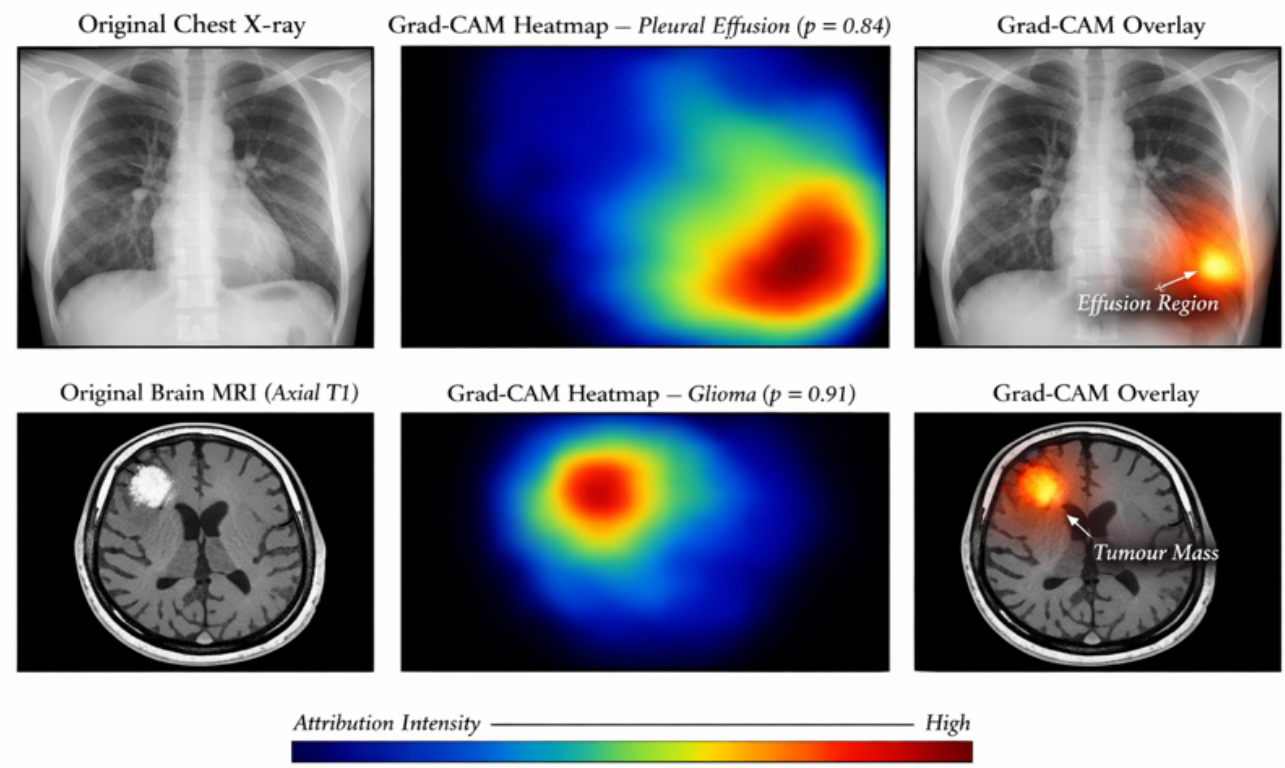

Fig. 3. Grad-CAM Heatmap

deployment. Performance evaluation disaggregated by age, sex, and race is required before deployment, as the CheXpert dataset has documented demographic imbalances. The contradiction detection rule library and reference table must be reviewed for demographic sensitivity, as reference ranges derived from predominantly one demographic group may not reflect normal variation in others.

XI. Conclusion

This paper presented ScanWise, a multimodal clinical decision support system integrating chest X-ray classification, brain MRI tumour classification, NLP-based biomarker extraction, decision-level fusion, and contradiction detection into a unified pipeline. The system is built on established pretrained architectures fine-tuned for their respective imaging tasks and a formally defined weighted fusion model combining imaging probability vectors, biomarker deviation vectors, and text embeddings into a composite risk score.

The primary contributions are the multimodal fusion formulation, the clinical contradiction detection rule module, the Grad-CAM-based spatial explainability integration, and the template-based structured clinical report generation engine. Unlike prior multimodal medical AI systems focused primarily on report generation or image-text alignment, ScanWise targets the specific clinical workflow requirement of cross-modal consistency and unified risk communication.

Acknowledgment

The authors would like to express their gratitude to Mr. Praveen Giridhar Pawaskar, Assistant Professor, School of Computer Science and Engineering, Presidency University, for his guidance and support, and his feedback during the entire process of this research. The authors also Presidency University for the support of the resources and the infrastructure needed to complete this work.

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